ERP PLATFORM

Software Requirements Specification

Procurement · Inventory · Finance

Version	1.0
Date	April 2026
Status	Draft
Prepared For	Internal Development Team
Tech Stack	React + Node.js + PostgreSQL
Cloud	Microsoft Azure

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1. Executive Summary

This document defines the Software Requirements Specification (SRS) for a modular, cloud-native Enterprise Resource Planning (ERP) platform. The system is designed to support the end-to-end business operations of distribution and trading companies through three core, independently configurable modules: Procurement, Inventory, and Finance.

The platform is built on a modern technology stack — React (frontend), Node.js (backend), and PostgreSQL (database) — and deployed on Microsoft Azure, enabling enterprise-grade scalability, availability, and security. The architecture supports role-based access control (RBAC), workflow automation, real-time dashboards, and a rich transactional UI similar to industry-leading ERP systems such as Azentio ORION.

The intended users include system administrators, procurement managers, inventory managers, finance officers, sales managers, and executive users, each accessing a tailored view of the platform.

2. System Overview

2.1 Purpose

The ERP platform centralizes business operations, replaces siloed tools, reduces manual effort, and provides real-time visibility into procurement, inventory, and financial health. Each module can be independently enabled or disabled by system administrators, making the platform deployable for companies of varying sizes and operational requirements.

2.2 Scope

The system covers the following functional areas:

- Supplier management, purchase requisitions, enquiries, quotations, and purchase orders (Procurement)
- Warehouse management, stock tracking, goods receipt, issue, and transfer (Inventory)
- Accounts payable, accounts receivable, general ledger, cost centers, and budgets (Finance)
- Role-based access control with configurable user profiles
- Workflow engine for approval routing and escalation
- Real-time KPI dashboards and standard reports
- Audit trails and document versioning
- Azure-hosted infrastructure with CI/CD pipelines

2.3 Stakeholders

Stakeholder	Role	Primary Interest
Business Owner	Executive Sponsor	ROI, operational efficiency, reporting
IT / DevOps	System Administrator	Deployment, uptime, security, module config
Procurement Manager	Module User	Purchase workflows, supplier management
Inventory Manager	Module User	Stock control, warehouse operations
Finance Manager	Module User	AP/AR, GL, budgets, financial reporting
Sales Manager	Module User	Demand visibility, order status
Warehouse Staff	Operational User	Goods receipt, picking, stock movement

3. Module Configuration System

Each of the three core modules (Procurement, Inventory, Finance) can be independently toggled on or off at the tenant/company level by the System Administrator. When a module is disabled, all associated menu items, API endpoints, and data-entry screens are hidden. Cross-module dependencies are enforced — for example, Finance cannot be enabled without Inventory, as GRN posting triggers journal entries.

3.1 Module Dependency Matrix

Module	Depends On	Can Run Standalone
Procurement	Inventory (for GRN)	Yes (PO only mode)
Inventory	None	Yes
Finance	Inventory + Procurement	No

3.2 Configuration Parameters

The following settings are configurable per company/tenant:

- Enable/disable each module (Procurement, Inventory, Finance)
- Set default currency and exchange rate mode (manual / auto-fetch)
- Define financial year start month
- Configure document numbering sequences per module
- Set multi-location / multi-warehouse mode
- Enable/disable TDS (Tax Deducted at Source) applicability per supplier
- Configure approval workflow levels (1-tier, 2-tier, 3-tier)
- Set notification preferences (email / in-app / both)

4. User Roles and Access Control

The platform implements Role-Based Access Control (RBAC). Every screen, API endpoint, and action is protected by a combination of role, module enablement, and record-level permissions. Roles are assigned per user and can be scoped to specific locations or cost centers.

4.1 Standard Roles

Role	Modules Accessible	Key Permissions
System Admin	All	Full access: configure modules, manage users, view all reports, manage sequences
Procurement Manager	Procurement, Inventory (read)	Create/approve PRs, POs, manage suppliers, run procurement reports
Purchase Officer	Procurement	Create PRs, PEs, PQs; view POs; cannot approve above own limit
Inventory Manager	Inventory, Procurement (read)	Manage stock, warehouses, GRN, issue, transfer, adjust
Warehouse Staff	Inventory (limited)	Create GRN, stock issue, transfer; no delete/approve
Finance Manager	Finance, read-only on others	AP/AR, GL entries, budget management, full financial reports
Accounts Officer	Finance (limited)	Enter invoices, payments; cannot post GL directly
Sales Manager	Inventory (read), Finance (AR)	View stock levels, demand, customer invoices and aging
Executive / CEO	All (read-only)	View dashboards, KPIs, and executive reports only

4.2 Permission Levels

Each role is assigned a permission level per module:

- View — read-only access to records
- Create — can create new records (drafts)
- Edit — can edit own or assigned records before approval
- Approve — can approve documents within their authorization limit
- Delete — can delete draft documents (not approved/posted records)
- Void / Cancel — can void approved documents (requires higher privilege)
- Configure — can change system settings for that module

4.3 Authorization Limits

Procurement and Finance approvals enforce monetary authorization limits per user. For example, a Purchase Officer may approve requisitions up to USD 5,000, while a Procurement Manager approves up to USD 50,000, and amounts above that escalate to the CEO or Board-level role. Limits are configurable in the admin console.

5. Procurement Module

The Procurement module manages the full purchase-to-pay cycle: from identifying a need (Material Requisition) through supplier evaluation, quotation comparison, purchase order creation, goods receipt, and supplier invoice matching.

5.1 Sub-Modules and Transactions

Transaction	Document Code	Description
Material Requisition	MRL-XXXXXXXX	Internal request to purchase items; triggers approval workflow
Purchase Requisition	PRL-XXXXXXXX	Approved material requisition converted to purchase requisition
Purchase Enquiry	PEL-XXXXXXXX	RFQ sent to one or more suppliers
Purchase Quotation	PQL-XXXXXXXX	Supplier response captured and compared
Purchase Order	POL-XXXXXXXX	Confirmed order sent to supplier
Supplier Return	SRL-XXXXXXXX	Return of items to supplier with debit note

5.2 Supplier Management

5.2.1 Supplier Master

The supplier master record stores all supplier details required for procurement and payment. Key fields include:

- Supplier Code (auto-generated), Name, Short Name
- Control Account (links to Finance GL)
- Credit Limit Days and Credit Amount
- Parent Supplier (for group-level suppliers)
- Shipment Mode (Air, Sea, Land, NA)
- TDS Applicable flag and TDS Party designation
- Payment Terms (net 30, net 60, etc.)
- Bank Details (for payment processing)
- Product Supply categories
- Incoterms (EXW, FOB, CIF, DDP, etc.)
- Finance Group allocation
- Multi-currency support per supplier
- Custom fields section for company-specific attributes
- Location / branch assignment

5.2.2 Supplier Onboarding Workflow

New suppliers go through an onboarding workflow: Create Draft > Submit for Approval > Finance Review (bank/credit) > Activate. Inactive suppliers are flagged and cannot be used in new transactions.

5.3 Material Requisition (MRL)

Material Requisitions originate from department heads or warehouse staff when stock needs replenishment. Key features:

- Header: Doc Location, Doc Date, Charge Code (Cost Center), Delivery Required On, Remarks
- Line items: Item Code, Description, Grade 1, Grade 2, UOM, Free Stock (live from inventory), Requested Qty, Approved Qty, Approx. Purchase Price
- Status: Draft > Submitted > Approved / Rejected > Converted to PR
- Multi-level approval based on total value
- Visibility of Obsolete Stock, Inactive Stock, Dead Stock, Pending Issues, Pending LTO, Pending GRN on the header

5.4 Purchase Requisition (PRL)

A Purchase Requisition is either manually created or auto-generated from an approved MRL. It consolidates items to be purchased and routes them to the purchasing team.

- Line Items: Item, Grade, UOM, Free Stock, Requested Qty, Approved Qty, Approx. Purchase Price, Charge Code
- Delivery Schedule, A/C Details, Alternate Items, Item Status tracking
- Short Close capability for partially fulfilled lines
- Lead Time tracking per item/supplier
- Status: Draft > Approved > Enquiry Sent > PO Created > Closed

5.5 Purchase Enquiry (RFQ)

Purchase enquiries are sent to one or more suppliers for a given set of items. The system tracks responses and enables comparison.

- Link to PRL lines
- One enquiry can cover multiple suppliers
- Supplier response deadline tracking
- Auto-email dispatch to suppliers (configurable template)
- Status per supplier: Sent > Responded > Closed

5.6 Purchase Quotation (PQL)

Supplier quotations are entered against the enquiry. A comparison report is generated to aid supplier selection.

- Price, currency, lead time, validity date, payment terms per supplier
- Comparison of quotation worksheet
- Award to supplier: marks selected quotation and closes others
- History of all quotations per item

5.7 Purchase Order (POL)

Purchase Orders are created from approved PRLs or quotations. They represent the legally binding commitment to the supplier.

- PO types: Local, Import, Non-Stock, Trading, Asset
- Header: Supplier, Doc Date, Delivery Date, Currency, Exchange Rate, Payment Terms, Incoterms, Ship-to Address, Bill-to Address
- Lines: Item, Description, UOM, Qty, Unit Price, Discount, Tax, Net Amount

- Multi-line approval workflow
- PO amendment history
- Short close / partial close
- GRN status per line
- 3-way match: PO vs GRN vs Supplier Invoice

6. Inventory Module

The Inventory module manages all aspects of stock: receipt, storage, issue, transfer, and adjustment. It supports multi-location, multi-warehouse, and multi-bin configurations.

6.1 Item Master

- Item Code, Description, Short Description
- Item Category, Sub-Category
- UOM (primary and secondary)
- Grade 1 and Grade 2 (quality classification)
- Reorder Level, Reorder Qty, Min Stock, Max Stock
- Lead Time (days)
- Standard Cost, Average Cost, Last Purchase Price
- Shelf Life / Expiry Tracking flag
- Serial / Batch / Lot Tracking flag
- Item Status: Active, Inactive, Obsolete, Dead
- Supplier cross-reference (supplier item codes)
- Custom fields

6.2 Warehouse and Location Management

- Multi-warehouse support: each warehouse has a code, name, and location
- Bin / rack location management within warehouses
- Preferred bin assignment per item
- Transfer between warehouses and bins

6.3 Goods Receipt Note (GRN)

GRNs record the physical receipt of goods against a Purchase Order.

- Reference to PO and supplier
- Actual quantities received vs ordered
- Quality check status (Accepted, Rejected, Partial)
- Lot / batch / serial number capture
- Bin assignment at receipt
- Automatic stock balance update
- Triggers Finance posting (if Finance module enabled)

6.4 Stock Issue

Issues record the movement of stock out of the warehouse to a cost center or production area.

- Reference to Material Requisition or direct issue
- Issue Qty, UOM, Bin, Cost Center
- Average cost used for valuation
- Auto-reduce stock balance

6.5 Stock Transfer

Transfers move stock between warehouses or bins without changing ownership.

- Source Warehouse/Bin, Target Warehouse/Bin
- Item, Qty, UOM, Lot/Batch
- In-Transit status for inter-warehouse transfers

6.6 Stock Adjustment

- Positive and negative adjustments (variance from physical count)
- Reason codes (damaged, miscounted, expired, etc.)
- Approval required for adjustments above threshold
- Automatic journal entry in Finance

6.7 Physical Stock Count

- Cycle count and full count modes
- Freeze stock during count
- Variance report (book vs physical)
- Post adjustments after approval

6.8 Stock Valuation

- Weighted Average Cost (WAC) method
- FIFO method (configurable)
- Stock aging report
- Dead/Inactive/Obsolete stock classification rules (configurable thresholds)

7. Finance Module

The Finance module provides full accounting functionality including accounts payable, accounts receivable, general ledger, cost center accounting, and budget management.

7.1 Chart of Accounts

- Account Code, Description, Account Type (Asset, Liability, Equity, Revenue, Expense)
- Control accounts for debtors and creditors
- Account hierarchy (Group > Sub-Group > Account)
- Multi-currency accounts
- Active/Inactive flag

7.2 Accounts Payable (AP)

- Supplier Invoice registration against PO/GRN (3-way match)
- Invoice approval workflow
- Payment processing: cheque, bank transfer, cash
- Advance payment management
- Credit note / debit note processing
- Supplier aging report
- SOA (Statement of Account) generation
- TDS computation and certificate generation
- Payment run (batch payment scheduling)

7.3 Accounts Receivable (AR)

- Customer master with credit terms
- Sales invoice registration
- Receipt processing
- Customer aging report
- Customer SOA
- Credit limit monitoring

7.4 General Ledger (GL)

- Manual journal entries
- Recurring journal templates
- Period-end closing
- Trial balance
- Profit & Loss statement
- Balance sheet
- Multi-period comparisons

7.5 Cost Center Accounting

- Cost center hierarchy
- Cost allocation rules
- Cost center P&L

- All transactions tagged to a cost center

7.6 Budget Management

- Annual budget entry by account and cost center
- Monthly budget phasing
- Budget vs Actual report
- Over-budget alerting and approval control

8. UI / UX Requirements

The user interface mirrors the design language of enterprise ERP systems like Azentio ORION, featuring a dense, information-rich layout designed for power users.

8.1 Layout Structure

- Left sidebar navigation: collapsible, organized by module and sub-module
- Top navigation bar: app logo, document search, global actions, notifications, user profile
- Tab bar: multiple documents open simultaneously (like browser tabs)
- Toolbar: context-specific actions (Save, New, Copy, Delete, Navigate, Print)
- Key Info / Item Details tabs within document forms
- Sub-tabs for related sections (A/C Details, Delivery Schedule, Alternate Items, etc.)
- Grid/table views for line items with inline editing
- Dashboard home screen: My Work (Workflow tasks), 360 View, Portlets, KPIs

8.2 Navigation and Tabs

- Breadcrumb-style document reference shown in the top right (e.g., PRL-202600123 | 06/04/2026 | Approved)
- Navigation arrows (First, Previous, Next, Last) within record sets
- Refresh button
- Document status badge (Draft, Submitted, Approved, Closed, Voided)

8.3 Form Design

- Required fields marked with asterisk (*)
- Lookup search fields with magnifier icon
- Multi-language flag icons for name fields
- Inline grids with add/delete row controls
- Color-coded status tiles (Inactive Stock, Dead Stock, Pending GRN, etc.)
- Right-panel document value summary (Approx Net Value)
- Custom Fields accordion section

8.4 Dashboard and KPIs

- Task View: Pending Tasks with traffic-light status (Within Target, Approaching, Beyond Target)
- Status View: Document statuses summary
- Integrated View: Cross-module operational view
- KPI portlets: configurable per role
- My Calendar: task and deadline calendar

9. Database Design

The database is PostgreSQL. Schema is organized into logical schemas per module for clean separation. All tables include standard audit fields: created_at, updated_at, created_by, updated_by, is_deleted (soft delete).

9.1 Core / Shared Schema (core)

Table	Key Fields	Purpose
companies	id, code, name, base_currency, modules_enabled	Multi-tenant company/tenant registry
users	id, company_id, email, password_hash, role_id, is_active	Application users
roles	id, name, description	RBAC roles definition
permissions	id, role_id, module, resource, action	Fine-grained permission matrix
doc_sequences	id, company_id, module, doc_type, prefix, next_no	Document number sequences
locations	id, company_id, code, name, type, address	Physical locations / branches
currencies	id, code, name, symbol	Currency master
exchange_rates	id, currency_id, rate_date, rate	Daily exchange rates
workflow_configs	id, company_id, module, doc_type, levels, escalation_hours	Approval workflow setup
audit_logs	id, table_name, record_id, user_id, action, old_val, new_val, ts	Full audit trail

9.2 Procurement Schema (procurement)

Table	Key Fields
suppliers	id, company_id, code, name, short_name, control_account_id, credit_days, credit_amount, parent_supplier_id, shipment_mode, is_active
supplier_bank_details	id, supplier_id, bank_name, account_no, iban, swift
supplier_contacts	id, supplier_id, name, designation, email, phone

Table	Key Fields
material_requisitions	id, company_id, doc_no, doc_date, location_id, charge_code_id, delivery_date, remarks, status, created_by
mrl_lines	id, mrl_id, item_id, grade1, grade2, uom_id, free_stock, req_qty, approved_qty, approx_price
purchase_requisitions	id, company_id, doc_no, doc_date, location_id, charge_code_id, delivery_date, remarks, status, mrl_id
prl_lines	id, prl_id, item_id, grade1, grade2, uom_id, free_stock, req_qty, approved_qty, approx_price, charge_code_id
purchase_enquiries	id, company_id, doc_no, doc_date, prl_id, status
purchase_quotations	id, company_id, doc_no, supplier_id, enquiry_id, validity_date, currency_id, status
purchase_orders	id, company_id, doc_no, doc_date, supplier_id, currency_id, exchange_rate, payment_terms, incoterms, delivery_date, status
po_lines	id, po_id, item_id, uom_id, ordered_qty, received_qty, unit_price, discount_pct, tax_pct, net_amount, charge_code_id

9.3 Inventory Schema (inventory)

Table	Key Fields
items	id, company_id, code, description, category_id, uom_id, reorder_level, reorder_qty, standard_cost, tracking_type, status
warehouses	id, company_id, code, name, location_id, is_active
bins	id, warehouse_id, code, name, capacity
stock_balances	id, item_id, warehouse_id, bin_id, qty_on_hand, qty_reserved, avg_cost
grn_headers	id, company_id, doc_no, po_id, supplier_id, warehouse_id, doc_date, status
grn_lines	id, grn_id, item_id, po_line_id, received_qty, accepted_qty, rejected_qty, bin_id, lot_no
stock_issues	id, company_id, doc_no, doc_date, warehouse_id, charge_code_id, mrl_id, status
stock_issue_lines	id, issue_id, item_id, bin_id, issued_qty, uom_id, avg_cost
stock_transfers	id, company_id, doc_no, doc_date, from_warehouse_id, to_warehouse_id, status
stock_adjustments	id, company_id, doc_no, doc_date, warehouse_id, reason_id, status, approved_by

9.4 Finance Schema (finance)

Table	Key Fields
gl_accounts	id, company_id, code, name, account_type, parent_id, is_control, currency_id, is_active
cost_centers	id, company_id, code, name, parent_id, budget_holder_id
journal_entries	id, company_id, doc_no, entry_date, description, status, source_module, source_doc_id
journal_lines	id, journal_id, account_id, cost_center_id, debit, credit, currency_id, fx_rate
ap_invoices	id, company_id, doc_no, supplier_id, po_id, grn_id, invoice_no, invoice_date, amount, tax, total, status
ap_payments	id, company_id, doc_no, supplier_id, payment_date, amount, payment_method, bank_account_id
ar_invoices	id, company_id, doc_no, customer_id, invoice_date, amount, tax, total, status
ar_receipts	id, company_id, doc_no, customer_id, receipt_date, amount, payment_method
budgets	id, company_id, fiscal_year, account_id, cost_center_id, annual_amount
budget_periods	id, budget_id, period_month, period_year, budgeted_amount, actual_amount

10. API Architecture

The backend exposes a RESTful API built with Node.js (Express.js / Fastify). All endpoints are versioned (/api/v1/...) and secured with JWT Bearer token authentication. API documentation is auto-generated with Swagger/OpenAPI 3.0.

10.1 Authentication Endpoints

Method	Endpoint	Description
POST	/api/v1/auth/login	Authenticate user, return JWT access token + refresh token
POST	/api/v1/auth/refresh	Exchange refresh token for new access token
POST	/api/v1/auth/logout	Invalidate refresh token
GET	/api/v1/auth/me	Get current user profile and permissions

10.2 Procurement API (sample)

Method	Endpoint	Description
GET	/api/v1/procurement/suppliers	List suppliers with filters (active, location, search)
POST	/api/v1/procurement/suppliers	Create supplier
PUT	/api/v1/procurement/suppliers/:id	Update supplier
GET	/api/v1/procurement/mrl	List Material Requisitions
POST	/api/v1/procurement/mrl	Create MRL
POST	/api/v1/procurement/mrl/:id/submit	Submit MRL for approval
POST	/api/v1/procurement/mrl/:id/approve	Approve / Reject MRL
POST	/api/v1/procurement/mrl/:id/convert-to-pr	Convert MRL to Purchase Requisition
GET	/api/v1/procurement/prl	List Purchase Requisitions
POST	/api/v1/procurement/po	Create Purchase Order
GET	/api/v1/procurement/po/:id/status	PO fulfilment status with GRN details

10.3 Inventory API (sample)

Method	Endpoint	Description
GET	/api/v1/inventory/items	List items with stock balances
GET	/api/v1/inventory/stock-balance	Stock balance by item/warehouse/bin
POST	/api/v1/inventory/grn	Create GRN
POST	/api/v1/inventory/grn/:id/post	Post GRN (updates stock balance)

Method	Endpoint	Description
POST	/api/v1/inventory/issue	Create stock issue
POST	/api/v1/inventory/transfer	Create stock transfer
POST	/api/v1/inventory/adjustment	Create stock adjustment (requires approval)

10.4 Finance API (sample)

Method	Endpoint	Description
GET	/api/v1/finance/accounts	Chart of accounts
POST	/api/v1/finance/journal	Create manual journal entry
GET	/api/v1/finance/trial-balance	Trial balance for period
GET	/api/v1/finance/ap/invoices	Supplier invoices with aging
POST	/api/v1/finance/ap/payment	Record supplier payment
GET	/api/v1/finance/reports/pnl	P&L statement for period

11. Technology Stack

11.1 Frontend

Technology	Version / Package	Purpose
React	18.x	Core UI framework
TypeScript	5.x	Type safety across frontend
Tailwind CSS	3.x	Utility-first styling
React Query (TanStack)	5.x	Server state management and caching
Zustand	4.x	Client state management
React Hook Form	7.x	Form state and validation
AG Grid	31.x	High-performance data grids for line items
Recharts / ApexCharts	Latest	Dashboard charts and KPIs
React Router	6.x	Client-side routing
Axios	1.x	HTTP client
Vite	5.x	Build tool and dev server

11.2 Backend

Technology	Version / Package	Purpose
Node.js	20.x LTS	Runtime
TypeScript	5.x	Type safety across backend
Fastify	4.x	High-performance HTTP framework
Prisma ORM	5.x	Database access, migrations, type-safe queries
PostgreSQL	16.x	Primary relational database
Redis	7.x	Session cache, job queues, rate limiting
JWT + Passport.js	Latest	Authentication and RBAC middleware
BullMQ	5.x	Background job queue (emails, reports)
Nodemailer / Azure Comm.	Latest	Email notifications
Swagger / Fastify Swagger	Latest	API documentation
Jest + Supertest	Latest	Unit and integration testing

11.3 Azure Infrastructure

Azure Service	Tier	Purpose
Azure App Service / AKS	P2v3 / Standard	Host Node.js backend (containerized)
Azure Static Web Apps	Standard	Host React frontend with global CDN
Azure Database for PostgreSQL	General Purpose, 4 vCores	Managed PostgreSQL with HA and backups
Azure Cache for Redis	C1 Standard	Session cache and job queues
Azure Blob Storage	LRS	Document attachments, report exports
Azure API Management	Developer / Standard	API gateway, rate limiting, monitoring
Azure Active Directory B2C	Included	SSO and identity management
Azure Key Vault	Standard	Secrets, connection strings, encryption keys
Azure Monitor + App Insights	Pay-as-you-go	Logging, tracing, alerting
Azure DevOps / GitHub Actions	Basic + free minutes	CI/CD pipelines
Azure Front Door	Standard	WAF, HTTPS, global load balancing

12. Non-Functional Requirements

12.1 Performance

- API response time: < 300ms for 95th percentile under normal load
- Page load time: < 2 seconds (Time to Interactive) on a standard corporate network
- Grid rendering: support for 10,000+ line items with virtual scrolling (AG Grid)
- Concurrent users: support 500 simultaneous active users per tenant

12.2 Scalability

- Horizontal scaling: stateless backend allows adding instances behind Azure Load Balancer
- Database connection pooling via PgBouncer
- Multi-tenant architecture with company_id isolation on all tables
- Async report generation via BullMQ job queue to avoid request timeout

12.3 Security

- All data in transit encrypted via TLS 1.2+ (enforced by Azure Front Door)
- Data at rest encrypted (Azure managed keys)
- JWT tokens expire in 15 minutes; refresh tokens in 7 days (stored in HttpOnly cookie)
- RBAC enforced both at API middleware level and database row level
- SQL injection prevention via Prisma parameterized queries
- XSS prevention: React default escaping + Content-Security-Policy headers
- Rate limiting: 100 requests/minute per IP via API Management
- Full audit log for all create/update/delete operations
- Secrets stored in Azure Key Vault (no secrets in code or config files)

12.4 Availability

- Target SLA: 99.9% uptime (Azure App Service SLA + PostgreSQL HA)
- Automated database backups: daily full + continuous WAL archiving (7-day retention)
- Zero-downtime deployments via blue/green deployment in Azure App Service
- Health check endpoint: /api/v1/health (monitored by Azure Monitor)

12.5 Maintainability

- Monorepo structure (frontend + backend in same repo via npm workspaces or Turborepo)
- TypeScript across the full stack
- Prisma migration files committed to git
- Minimum 80% unit test coverage on business logic
- OpenAPI spec auto-generated and published to developer portal

13. Reports and Queries

13.1 Procurement Reports

- Material Requisition Status Report
- Purchase Requisition Status Report
- Purchase Order Status Report (open, partial, closed)
- Purchase Order History by Supplier/Item
- Procurement Tracking (PR-to-PO cycle time)
- Pending Purchase Requisitions
- Lead Time Variance Report
- Purchase Price Comparison (vs budget, vs last purchase)
- Schedule vs Actual Supply
- Comparison of Quotations

13.2 Inventory Reports

- Stock Balance Report (by item, warehouse, bin)
- Stock Aging Report
- Dead / Inactive / Obsolete Stock Report
- GRN Summary Report
- Stock Movement Report
- Reorder Level Report
- Stock Valuation Report (WAC / FIFO)
- Physical Count Variance Report

13.3 Finance Reports

- Trial Balance
- Profit & Loss Statement
- Balance Sheet
- Supplier Aging Report
- Customer Aging Report
- Budget vs Actual
- Cost Center P&L
- Cash Flow Statement
- TDS Summary Report

14. Integration Requirements

The platform is designed to integrate with external systems via REST APIs and webhooks.

Integration	Type	Details
Email (SMTP / Azure)	Outbound	Workflow notifications, supplier RFQs, payment confirmations
Azure Active Directory	SSO	Enterprise SSO login for corporate users
Currency Exchange API	Inbound REST	Auto-fetch daily exchange rates (e.g., Open Exchange Rates)
Document Storage	Azure Blob	Attachment upload/download for all documents
Report Export	PDF / Excel	All reports exportable via server-side rendering (Puppeteer / ExcelJS)
Webhook Outbound	REST Webhook	Notify external systems on PO approval, GRN posting, etc.

15. Development Phases

Phase	Name	Duration	Key Deliverables
1	Foundation	2 weeks	Monorepo setup, DB schema, auth, RBAC, module config
2	Procurement Backend	3 weeks	Supplier, MRL, PRL, PE, PQ, PO APIs + workflow engine
3	Procurement Frontend	3 weeks	All procurement screens, document forms, grids
4	Inventory Backend	2 weeks	Item, Warehouse, GRN, Issue, Transfer, Adjustment APIs
5	Inventory Frontend	2 weeks	All inventory screens, stock balance views
6	Finance Backend	3 weeks	GL, AP, AR, Budget APIs + auto-posting from Inv/Proc
7	Finance Frontend	3 weeks	All finance screens, journal entry, reports
8	Dashboard & Reports	2 weeks	KPI dashboards, all standard reports, export
9	Testing & QA	2 weeks	Integration tests, UAT, performance tuning
10	Azure Deployment	1 week	Infrastructure as Code (Bicep), CI/CD, go-live

Total estimated development timeline: 23 weeks (approximately 6 months) with a team of 3-4 full-stack developers.

Appendix A: Glossary

Term	Definition
MRL	Material Requisition — internal request to procure items
PRL	Purchase Requisition — formal request to purchasing department
PE / RFQ	Purchase Enquiry / Request for Quotation — sent to suppliers
PQL	Purchase Quotation — supplier price response to RFQ
PO	Purchase Order — legally binding purchase commitment
GRN	Goods Receipt Note — records physical receipt of goods
AP	Accounts Payable — money owed to suppliers
AR	Accounts Receivable — money owed by customers
GL	General Ledger — central accounting record
WAC	Weighted Average Cost — stock valuation method
RBAC	Role-Based Access Control — permissions system
TDS	Tax Deducted at Source — withholding tax
SOA	Statement of Account — supplier/customer balance summary
LTO	Local Transfer Order — inter-warehouse stock transfer